

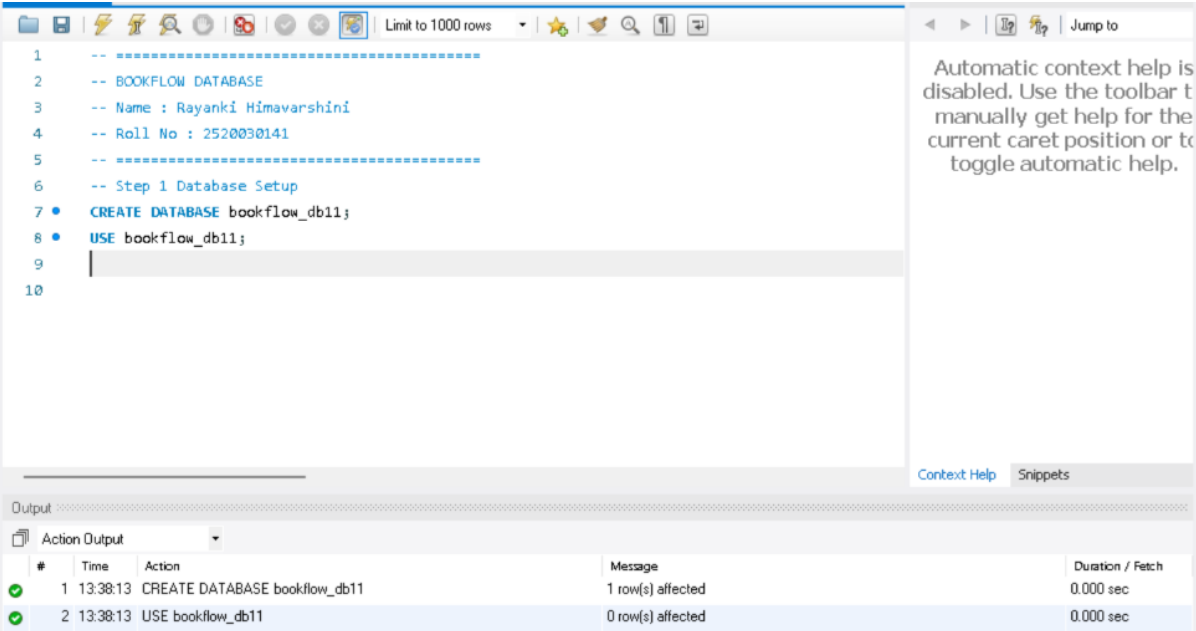
WEEK – 1

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Week 1:

A local library is currently managing its books and members using a physical notebook, which has resulted in several operational problems such as misplaced books, difficulty tracking borrowed items, and inaccurate member records. To modernize its operations, the library plans to develop a database system called **BookFlow**. As a database developer, your task is to create the SQL foundation for this system. First, create a new database named **bookflow_db**. Then, design two tables: a **Books** table containing the columns **book_id**, **title**, **isbn**, and **published_year**, and a **Members** table containing the columns **member_id**, **full_name**, and **email**. Ensure data integrity by setting **book_id** and **member_id** as **PRIMARY KEYS**, applying a **NOT NULL** constraint to the **title** column so that every book must have a name, enforcing **UNIQUE** constraints on both **isbn** and **email** to prevent duplicate entries, and adding a **CHECK** constraint to ensure that the **published_year** is not in the future (for example, it must be less than 2027). Finally, insert three sample records into both the **Books** and **Members** tables to verify that the database structure and constraints are functioning correctly. This exercise will help you develop skills in schema definition, data integrity enforcement, and the use of SQL Data Definition Language (DDL) commands such as **CREATE DATABASE**, **CREATE TABLE**, and **INSERT** statements.

Step 1: Create Database:



The screenshot displays a SQL IDE interface. The main editor window contains the following SQL script:

```
1  -- =====
2  -- BOOKFLOW DATABASE
3  -- Name : Rayanki Himavarshini
4  -- Roll No : 2520030141
5  -- =====
6  -- Step 1 Database Setup
7  • CREATE DATABASE bookflow_db11;
8  • USE bookflow_db11;
9
10
```

On the right side, a context help panel is visible with the text: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

At the bottom, the "Output" window is open, showing the "Action Output" tab. It contains a table with the following data:

#	Time	Action	Message	Duration / Fetch
✓ 1	13:38:13	CREATE DATABASE bookflow_db11	1 row(s) affected	0.000 sec
✓ 2	13:38:13	USE bookflow_db11	0 row(s) affected	0.000 sec

Step 2: Create the Books Table:

SQL File 1* x

Limit to 1000 rows

```

2  -- BOOKFLOW DATABASE
3  -- Name : Rayanki Himavarshini
4  -- Roll No : 2520030141
5  -- =====
6  -- Step 1 Database Setup
7  • CREATE DATABASE bookflow_db11;
8  • USE bookflow_db11;
9  -- Step 2 Table Creation (DDL) with Constraints
10 • CREATE TABLE books (
11     book_id INT AUTO_INCREMENT PRIMARY KEY,
12     title VARCHAR(255) NOT NULL,
13     isbn VARCHAR(13) NOT NULL UNIQUE,
14     published_year INT,
15     CONSTRAINT chk_published_year CHECK (published_year < 2027)
16 );
17

```

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	13:38:13	USE bookflow_db11	0 row(s) affected	0.000 sec
3	13:40:27	CREATE DATABASE bookflow_db11	Error Code: 1007. Can't create database 'bookflow_db11'; database exists	0.000 sec
4	13:41:04	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY K...	0 row(s) affected	0.015 sec

```

16 );
17 -- Create the members table
18 • CREATE TABLE members (
19     member_id INT AUTO_INCREMENT PRIMARY KEY,
20     full_name VARCHAR(100) NOT NULL,
21     email VARCHAR(150) NOT NULL UNIQUE
22 );
23
24

```

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	13:40:27	CREATE DATABASE bookflow_db11	Error Code: 1007. Can't create database 'bookflow_db11'; database exists	0.000 sec
4	13:41:04	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY K...	0 row(s) affected	0.015 sec
5	13:42:29	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIM...	0 row(s) affected	0.047 sec

```

23 • DESCRIBE books;
24

```

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Result Grid

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto
title	varchar(255)	NO		NULL	
isbn	varchar(13)	NO	UNI	NULL	
published_year	int	YES		NULL	

Result 4 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	13:42:29	CREATE TABLE members (member_i...	0 row(s) affected	0.047 sec
6	13:45:00	CREATE DATABASE bookflow_db11	Error Code: 1007. Can't create databas...	0.000 sec
7	13:45:03	DESCRIBE books	4 row(s) returned	0.015 sec / 0.000 sec

23 DESCRIBE books;
24 DESCRIBE members;
25

current caret position or toggle automatic help.

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_inc
full_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	

Result 5 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	13:45:00	CREATE DATABASE bookflow_db11	Error Code: 1007. Can't create databas...	0.000 sec
7	13:45:03	DESCRIBE books	4 row(s) returned	0.015 sec / 0.000 sec
8	13:47:05	DESCRIBE members	3 row(s) returned	0.000 sec / 0.000 sec

Step 3 Initial Seeding (Valid Data)

```

6 INSERT INTO books (title, isbn, published_year) \
7 ('The Alchemist', '9780061122415', 1988),
8 ('Clean Code', '9780132350884', 2008),
9 ('Atomic Habits', '9780735211292', 2018);

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	13:45:03	DESCRIBE books	4 row(s) returned	0.015 sec / 0.000 sec
8	13:47:05	DESCRIBE members	3 row(s) returned	0.000 sec / 0.000 sec
9	13:49:01	INSERT INTO books (title, isbn, publis...	3 row(s) affected Records: 3 Duplicate...	0.016 sec

Step 4 Insert 3 members

```

25 -- Step 3 Initial Seeding (Valid Data)
26 INSERT INTO books (title, isbn, published_year) \
27 ('The Alchemist', '9780061122415', 1988),
28 ('Clean Code', '9780132350884', 2008),
29 ('Atomic Habits', '9780735211292', 2018);
30 -- Insert 3 members
31 INSERT INTO members (full_name, email) VALUES
32 ('himavarshini', 'Hima@example.com'),
33 ('sutte alekhya', 'Alekhya@example.com'),
34 ('Ravi Verma', 'ravi.verma@example.com');
35
36

```

toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	13:49:01	INSERT INTO books (title, isbn, publis...	3 row(s) affected Records: 3 Duplicate...	0.016 sec
10	13:51:26	CREATE DATABASE bookflow_db11	Error Code: 1007. Can't create databas...	0.000 sec
11	13:51:28	INSERT INTO members (full_name, e...	3 row(s) affected Records: 3 Duplicate...	0.016 sec
12	13:51:31	INSERT INTO members (full_name, e...	Error Code: 1062. Duplicate entry 'Him...	0.000 sec

35 • `SELECT * FROM books;`
36

Result Grid

	book_id	title	isbn	published_year
▶	1	The Alchemist	9780061122415	1988
	2	Clean Code	9780132350884	2008
	3	Atomic Habits	9780735211292	2018
*	NULL	NULL	NULL	NULL

books 6

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✖	13 13:51:43	INSERT INTO members (full_name, e...	Error Code: 1062. Duplicate entry 'Him...	0.000 sec
✖	14 13:53:04	CREATE DATABASE bookflow_db11	Error Code: 1007. Can't create databas...	0.000 sec
✔	15 13:53:07	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec

current caret position or
toggle automatic help.

35 • `SELECT * FROM books;`
36 • `SELECT * FROM members;`
37

Result Grid

	member_id	full_name	email
▶	1	hima varshini	Hima@example.com
	2	sutte alekhya	Alekhya@example.com
	3	Ravi Verma	ravi.verma@example.com
*	NULL	NULL	NULL

members 7

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✖	14 13:53:04	CREATE DATABASE bookflow_db11	Error Code: 1007. Can't create databas...	0.000 sec
✔	15 13:53:07	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec
✔	16 13:53:38	SELECT * FROM members LIMIT 0, 1...	3 row(s) returned	0.000 sec / 0.000 sec

current caret position or
toggle automatic help.

Step 4 Constraint Tests (These MUST Fail)

```
7  -- Step 4 Constraint Tests (These MUST Fail)
8  • INSERT INTO books (title, isbn, published_year)
9    VALUES ('Fake Copy', '9780061122415', 2000);
10
11
12
13
14
15
16
17
18
19
```

Context Help Snippets

Output

#	Time	Action	Message	Duration / Fetch
16	13:53:38	SELECT * FROM members LIMIT 0,1...	3 row(s) returned	0.000 sec / 0.000 sec
17	13:55:50	CREATE DATABASE bookflow_db11	Error Code: 1007. Can't create databas...	0.000 sec
18	13:55:52	INSERT INTO books (title, isbn, publis...	Error Code: 1062. Duplicate entry '978...	0.000 sec

NULL title (NOT NULL violation)

```
34 ('Ravi Verma', 'ravi.verma@example.com');
35 • SELECT * FROM books;
36 • SELECT * FROM members;
37 -- Step 4 Constraint Tests (These MUST Fail)
38 • INSERT INTO books (title, isbn, published_year)
39   VALUES ('Fake Copy', '9780061122415', 2000);
40 -- NULL title (NOT NULL violation)
41 • INSERT INTO books (title, isbn, published_year)
42   VALUES (NULL, '99999999999999', 2010);
43
44
45
46
47
48
49
```

Context Help Snippets

Output

#	Time	Action	Message	Duration / Fetch
17	13:55:50	CREATE DATABASE bookflow_db11	Error Code: 1007. Can't create databas...	0.000 sec
18	13:55:52	INSERT INTO books (title, isbn, publis...	Error Code: 1062. Duplicate entry '978...	0.000 sec
19	14:00:37	INSERT INTO books (title, isbn, publis...	Error Code: 1048. Column 'title' cannot ...	0.000 sec